

Computer Science and Full-Stack Engineering Exam Bank

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1. What is the primary communication protocol used by Socket.IO for low-latency duplex transmission?

- A) HTTP Long Polling
- B) WebSocket
- C) SMTP Relay
- D) FTP Stream

Answer: B

Note: WebSocket provides full-duplex communication channels over a single TCP connection.

2. Which CSS property creates modern glassmorphism blur effects on background layers?

- A) filter: blur()
- B) backdrop-filter: blur()
- C) box-shadow: inset
- D) opacity: 0.5

Answer: B

Note: backdrop-filter applies graphical effects like blurring to the area behind an element.

3. In a Kahoot-style scoring algorithm, what factor increases point awards for correct answers?

- A) Player level
- B) Speed of answer submission
- C) Screen resolution
- D) Number of previous attempts

Answer: B

Note: Quicker response times yield maximum points up to 1000 per question.

4. Which HTTP status code indicates a successful resource creation?

- A) 200 OK
- B) 201 Created
- C) 204 No Content
- D) 304 Not Modified

Answer: B

Note: HTTP 201 indicates that the request has succeeded and led to resource creation.

5. What is the function of the Event Loop in JavaScript runtimes?

- A) To compile C++ code into machine bytecode
- B) To monitor the call stack and dispatch tasks from callback queues
- C) To encrypt network sockets
- D) To allocate heap memory directly to GPU buffers

Answer: B

Note: The event loop enables non-blocking asynchronous execution in single-threaded JavaScript.

6. Which data format is standard for transmitting structured data in web applications?

- A) XML-RPC
- B) JSON
- C) YAML-Binary
- D) INI Configuration

Answer: B

Note: JSON (JavaScript Object Notation) is universally supported and lightweight.

7. What does the DOM stand for in web browsers?

- A) Dynamic Object Map
- B) Document Object Model
- C) Digital Optimization Module
- D) Direct Operation Method

Answer: B

Note: The DOM represents HTML documents as a node tree.

8. Which cryptographic protocol secures HTTPS connections?

- A) TLS
- B) SSH
- C) PGP
- D) IPsec

Answer: A

Note: Transport Layer Security (TLS) encrypts web communications.

9. In relational databases, what does SQL stand for?

- A) Sequential Query Logic
- B) Structured Query Language
- C) Standard Quality Link
- D) System Quick Lookup

Answer: B

Note: SQL is the standard language for relational database management.

10. Which audio technology allows real-time sound synthesis in modern web browsers?

- A) Web Audio API
- B) Flash Audio Player
- C) QuickTime Plugin
- D) DirectSound ActiveX

Answer: A

Note: Web Audio API provides high-level modular audio synthesis and processing.

11. What is the time complexity of searching an element in a balanced Binary Search Tree?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n^2)$

Answer: B

Note: Balanced BST halves search space at each comparison step.

12. Which data structure operates on a First-In, First-Out (FIFO) basis?

- A) Stack
- B) Queue
- C) Heap
- D) Priority Array

Answer: B

Note: Queues process elements in arrival order.

13. Which port number is standard for unencrypted HTTP traffic?

- A) 21
- B) 80
- C) 443
- D) 8080

Answer: B

Note: Port 80 is the designated port for standard HTTP.

14. What does CPU stand for?

- A) Central Processing Unit
- B) Central Program Utility
- C) Core Performance User
- D) Computer Power Unit

Answer: A

Note: The CPU executes instruction cycles.

15. Which CSS layout mode provides two-dimensional grid-based layout capabilities?

- A) Flexbox
- B) CSS Grid
- C) Float Layout
- D) Absolute Positioning

Answer: B

Note: CSS Grid handles rows and columns simultaneously.